

Acoustic-Based Detection, Classification, and Localization of Impulsive Sounds

Jakub Svatoš Martin Maxa

Czech Technical University in Prague
Faculty of Electrical Engineering, Department of Measurement

⟨CONFERENCE NAME — TO BE CONFIRMED⟩

September 16, 2026

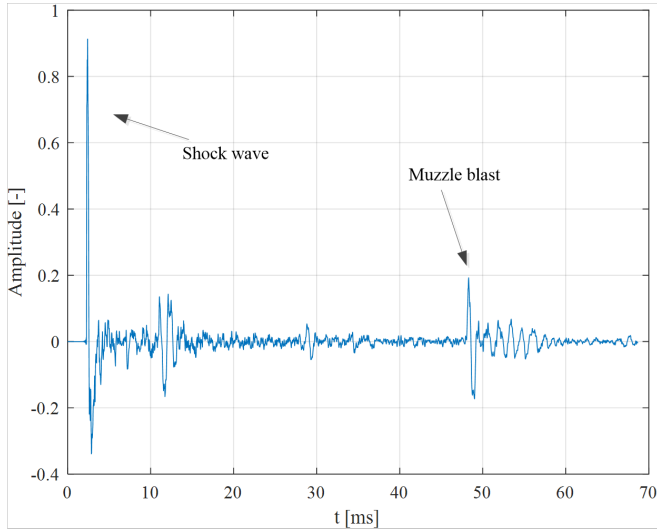
Why acoustic surveillance

- Shootings in public spaces are more frequent; global gun ownership is rising.
- Growing demand for autonomous security in communal areas — campuses, hospitals, parks, government facilities.
- Cameras, drones and security personnel give *some* protection, but fall short in high-risk situations with an active shooter.

What an acoustic system adds

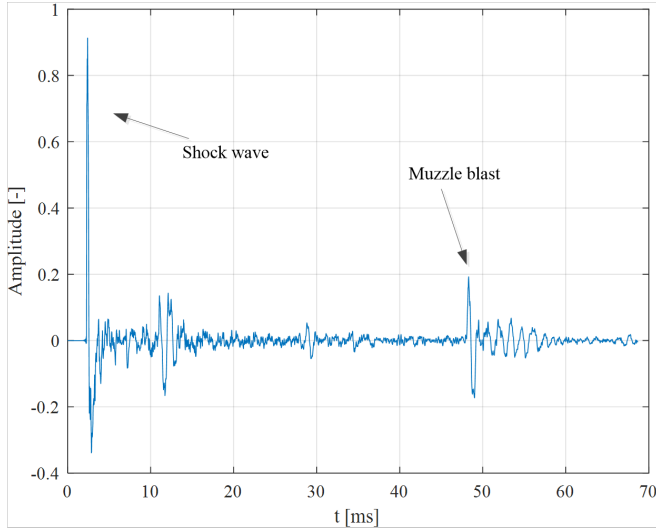
Identifies gunfire, pinpoints its source, and tracks an ongoing threat — faster detection and shorter response times for security forces.

The acoustic signature of a gunshot



- **Shock wave** (“N-wave”) — arrives *first*; exists only when the bullet is supersonic ($v > c$).

The acoustic signature of a gunshot



- **Shock wave** (“N-wave”) — arrives *first*; exists only when the bullet is supersonic ($v > c$).
- **Muzzle blast** — gases leaving the barrel, 1–3 ms; travels at the speed of sound, so it arrives ≈ 45 ms later.
- Everything between and after: **reflections**.

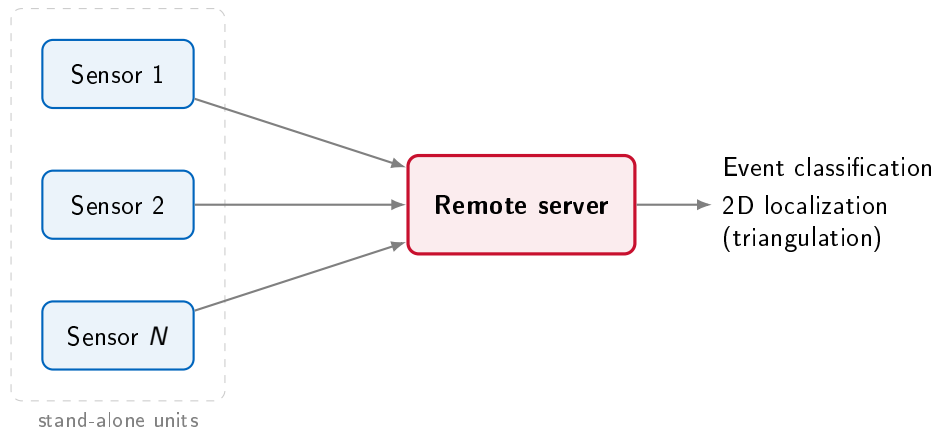
Why naive processing is not enough

- **Spectral analysis alone** often captures environmental noise rather than the gunshot itself.
- **Time-domain patterns** get distorted by nonlinear dispersion and obstacles.

Consequence

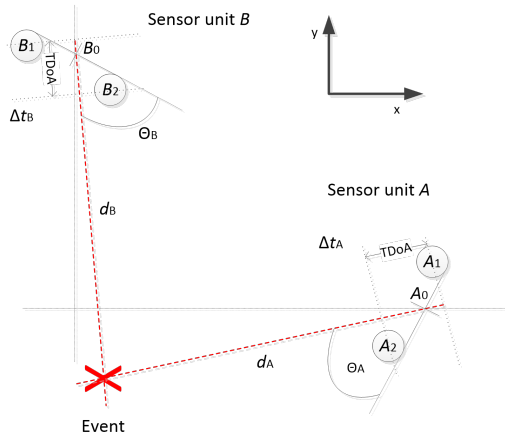
Effective feature extraction must integrate *both* time and frequency domain processing.

Proposed system architecture



- Each sensor continuously monitors its surroundings and transmits detected signals to the server for advanced processing.
- Multiple units give broader coverage and adaptability to environments.
- Tested with various firearm calibers and ammunition.

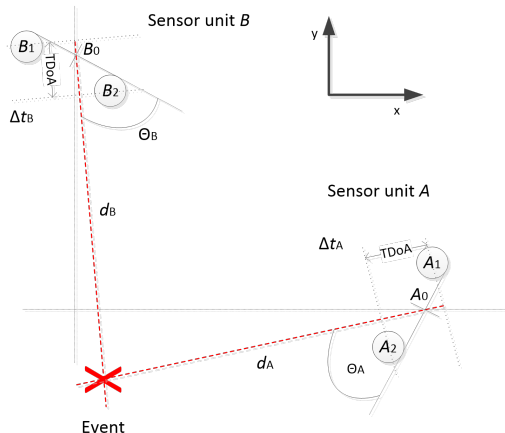
Locating the source



$$\Theta = \arcsin\left(\frac{c \Delta n}{d f_s}\right)$$

- Two MEMS microphones per unit ($d = 0.186$ m, $f_s = 44.1$ kHz) give one **bearing** Θ .

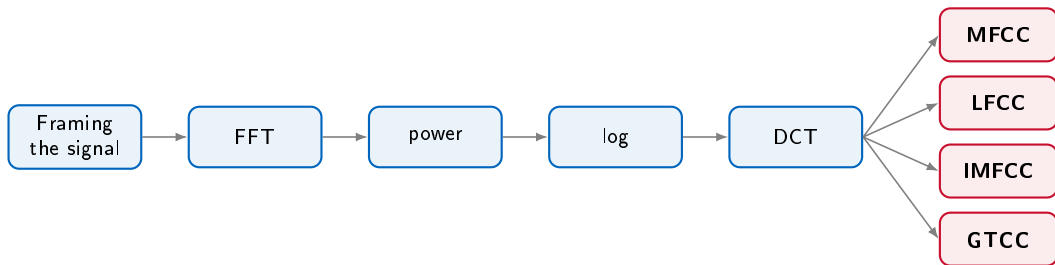
Locating the source



$$\Theta = \arcsin\left(\frac{c \Delta n}{d f_s}\right)$$

- Two MEMS microphones per unit ($d = 0.186$ m, $f_s = 44.1$ kHz) give one **bearing** Θ .
- Two units, two bearings — they **intersect** at the event.
- Measured: cross-correlation with parabolic interpolation, **MAE** 0.77° over 70 events (-45° to $+60^\circ$).

Feature extraction — cepstral coefficients



Block diagram of the feature extraction algorithms (Fig. 1 in the abstract); each branch applies its own filter bank before the DCT.

- MFCC was developed for other acoustic applications but works for gunshot recognition — validated at military shooting ranges.
- LFCC and IMFCC offer alternative filtering approaches.
- GFCC and GTCC derive from human auditory response models; **GTCC is more noise-resistant than MFCC.**

Classification results — 9 mm class

Frame	Extractor	ACC (%)		RLC (%)		MCC (–)	
		SVM	NN	SVM	NN	SVM	NN
15 ms	MFCC	83.90	85.59	80.65	67.74	0.62	0.62
	LFCC	83.90	83.90	80.65	64.52	0.62	0.57
	IMFCC	85.59	90.68	83.87	87.10	0.66	0.77
	GTCC	93.22	97.46	87.10	93.55	0.82	0.93
30 ms	MFCC	84.75	87.29	67.74	87.10	0.60	0.70
	LFCC	87.29	83.90	70.97	70.97	0.66	0.59
	IMFCC	85.59	92.37	67.74	80.65	0.62	0.80
	GTCC	97.46	98.31	90.32	93.55	0.93	0.96
50 ms	MFCC	84.75	93.22	74.19	90.32	0.61	0.83
	LFCC	88.98	86.44	77.42	77.42	0.71	0.66
	IMFCC	89.83	90.68	74.19	80.65	0.73	0.76
	GTCC	97.46	99.15	90.32	96.77	0.93	0.98

Table I in the abstract — example of classification results for a 9 mm class. Test partition: 118 events, 31 of them 9 mm.

What the numbers say

- MFCC, LFCC and IMFCC yield similar results.
- **GTCC consistently performs better** across every frame length and both classifiers.
- **NN outperforms SVM** — better at modelling complex patterns. The gap widens with longer frames.
- **Longer frames improve accuracy**, but start to capture environmental reflections, which can pull focus away from the gunshot itself.

Best configuration in the table

GTCC + NN at 50 ms — 99.15 % ACC, 96.77 % RLC, MCC 0.98.

Conclusion

- A distributed network of stand-alone acoustic sensors with a remote server detects, classifies and localizes impulsive sounds in 2D.
- Cepstral coefficients are an effective feature extractor for gunshot classification with both NN and SVM.
- GTCC beats Mel-frequency based coefficients; NN achieves slightly better classification success, especially with longer frames.
- Validated with various firearm calibers and ammunition.

Thank you — questions?

svatoja1@fel.cvut.cz maxamart@fel.cvut.cz

Backup — experimental setup and dataset

- Military shooting range, fixed sensor position, clear line of sight to every firing position.
- Four weapons: 9 mm, 5.56 NATO SD, 7.62 mm Tokarev, .22 — each with a single ammunition type.
- Plus a false-alarm class with gunshot-like impulsive patterns.
- 26 features per extractor; $\approx 75 / 25$ train–test split.
- NN: MATLAB Neural Fitting, Levenberg–Marquardt, 370 iterations.

Class		Train	Test
0	False alarms	86	21
1	9 mm	70	31
2	5.56 NATO SD	100	31
3	7.62 mm Tokarev	55	17
4	.22	60	18
Total		371	118

Svatoš & Holub, *Meas. Sci. Technol.* **35** (2024) 076122, Table 2.

Backup — bearing estimation methods

Method	MAE	Median	Max
Cross-correlation	0.84°	0.41°	7.61°
Cross-correlation + parabolic interp.	0.77°	0.56°	6.67°
Phase-based delay estimation (PBDE)	3.24°	1.80°	29.0°
PBDE + linear regression	3.91°	3.55°	12.15°

70 events, 9 reference angles from -45° to $+60^\circ$, $d = 0.186$ m, $f_s = 44.1$ kHz.

PBDE algorithm

1. FFT of both channels
2. select the relevant frequency bins
3. phase angle $\varphi = \arg(S_1 S_2^*)$
4. delay from the phase slope

How many units for 2D?

- ToA, 1 mic per unit — 3 units
- TDoA, 1 mic per unit — 4 units
- **TDoA, 2 mics per unit — 2 units**

Backup — other cepstral coefficients

- **LPCC** (Linear Prediction Cepstral Coefficients) and **PLPC** (Perceptual Linear Prediction Coefficients) — noise elimination, speech recognition.
- **GFCC** (Greenwood Function Cepstral Coefficients) — derived from human auditory response models, used for classifying animal sounds.
- **GTCC** (Gammatone-Frequency Cepstral Coefficients) — same family, greater resistance to noise than MFCC.

Backup — metrics

ACC Accuracy — share of correctly classified events.

RLC Recall, also known as *sensitivity* — share of actual gunshots of the class that were detected.

MCC Matthews Correlation Coefficient — balanced measure robust to class imbalance; +1 perfect, 0 random.

Definitions as used in the source study: Svatoš & Holub, *Meas. Sci. Technol.* **35** (2024) 076122, eqs. (1)–(3).